Francis **Banville**

Computational ecologist and data scientist with experience in One Health

francisbanville12@hotmail.com | francisbanville.github.io/ | FrancisBanville | francis-banville-890a91192

Profile

Research interests	computational ecology, biodiversity indicators, biostatistics, ecological networks, One Health
Programming	R, Julia, Git, Bash, Markdown, LaTeX
Specialized software	OpenRefine, QGIS, ArcGIS, Excel
Languages	French (native), English (fluent), Spanish (B2.2)

Education

In progress

Microprogram in Higher Education — *Montreal, CAN*

Université de Montréal — *Jan. 2025 - Dec. 2026*

- GPA: 4.225 (12 / 15 credits)

Completed

Ph.D. in Biological Sciences — *Montreal, CAN*

Université de Montréal — *Sept. 2019 - Dec. 2024*

- option General Biology
- advisors: Drs. Timothée Poisot and Dominique Gravel (Université de Sherbrooke)
- thesis: Towards a maximum entropy theory of food webs (classified as excellent, 84 research credits)
- GPA: 4.150/4.3 (6 course credits)

B.Sc. in Biological Sciences — *Montreal, CAN*

Université de Montréal — *Jan. 2016 - Apr. 2018*

- option Biodiversity, Ecology, and Evolution
- GPA: 4.198 / 4.3 (91 credits)

Interrupted

M.Sc. in Biological Sciences — *Montreal, CAN*

Université de Montréal — *Sept. 2018 - Aug. 2019*

- option Quantitative and Computational Biology (course-based M.Sc.)
- GPA: 4.217 / 4.3 (35 / 45 credits)

B.Sc. in Mathematics — *Montreal, CAN*

Université de Montréal — *Sept. 2013 - Aug. 2015*

- option Applied and Pure Mathematics
- GPA: 4.161 / 4.3 (33 / 90 credits)

Main professional experience

Postdoctoral fellow — *Montreal, CAN*

Université de Montréal — *Jan. 2025 - May 2026*

- assessing the usability of the KM-GBF indicators for monitoring One Health actions
- advisors: Drs. Timothée Poisot, Andrea Paz Velez, and Colin Carlson (Yale University)

Consultant — *Montreal, CAN*

Secretariat of the Convention on Biological Diversity (SCBD) — *Jan. 2025 - May 2026*

- developing indicators, metrics and progress measurements tools on biodiversity and health
- in collaboration with: the GEO BON One Health working group

Teaching assistant — *Montreal, CAN*

Université de Montréal, département de sciences biologiques — *Sept. 2020 - Dec. 2024*

- preparing course material, animating hands-on sessions, grading
- courses: biostatistics, population dynamics, research methods in biology, network theory

Grants and awards

Grants

2026	**Mobility grant**, Les Offices jeunesse internationaux du Québec (LOJIQ)	*$850*
2024	**Scholarship for end of PhD studies (5th year)**, Graduate and Postdoctoral Studies, U. de Montréal	*$8,000*
2023-24	**Fully funded fellowship**, Computational Biodiversity Science & Services program (BIOS^2)	*$10,500*
2019-24	**Mobility grants**, Computational Biodiversity Science & Services program (BIOS^2)	*$10,000*
2023	**Excellence Scholarship**, Fonds de bourses en sciences biologiques, U. de Montréal	*$1,500*
2023	**Perseverance Scholarship**, Département de sciences biologiques, U. de Montréal	*$2,000*
2019-23	**PhD Excellence Scholarship**, Institute for Data Valorization (IVADO)	*$100,000*
2019-21	**Fast-track Master's to PhD program**, Graduate and Postdoctoral Studies, U. de Montréal	*$14,000*
2019	**Research internship grant**, The Lapierre lab, Université de Montréal	*$6,000*
2018	**Research internship grant**, The Lapierre lab, Université de Montréal	*$8,400*
2017	**Undergraduate Student Research Awards**, Natural Sci. and Engr. Research Council of CAN (NSERC)	*$7,625*

Awards

2024	**Best presentation by a PhD student**, 34th Biology Symposium of the University of Montreal	*U. de Montréal*
2014-18	**Dean's Award of the Faculty of Arts and Sciences**, Distinction for academic excellence	*U. de Montréal*
2011	**Governor General's Academic Medal**, Distinction for the best overall average of the 2011 cohort	*Polybel high school*
2011	**Lieutenant Governor's Youth Medal**, Honorary medal for academic excellence and social involvement	*Polybel high school*

Scientific publications

Peer-reviewed articles

[14] Catchen, M.D., **Banville, F.**, Boutin, A.C., Brookson, C.B., Carlson, C.J., Dansereau, G., Gibb, R., Houle, M., Kaza, B., Robertson, H., Simons, D., Seifert, S.N., and Poisot, T. (2026). A protocol for biodiversity-informed wildlife disease surveillance. PLOS Ecosystems. https://doi.org/10.1371/journal.pesy.0000023 *2026*

[13] Soucy, C.P., **Banville, F.**, Beasley, E., Lefebvre, B., Gutiérrez-Marco, J.C., Poisot, T., and Cameron, C.B. (2026). A paleoecological trophic network model of the Early Ordovician Lower Fezouata Shale (Morocco) fossil fauna. iScience. https://doi.org/10.1016/j.isci.2026.116819 *2026*

[12] Bolduc, D., Dulude-de Broin, F., Bergeron, G., Villeneuve, C., Weiss-Blais, M., Couloigner, C., Dubourg, R., Fraser-Franco, M., **Banville, F.**, Moisan, L., Montiglio, P.-O., Fortin, D., Thibault, F., Allard, A., Gaudreau-Rousseau, C., Roca, I., Durand, A., Massé, C., Barreau, E., … Legagneux, P. (2025). Impersonating predators and prey to study trophic interactions through real-life simulations. Methods in Ecology and Evolution. http://doi.org/10.1111/2041-210x.70180 *2025*

[11] **Banville, F.**, Strydom, T., Blyth, P., Brimacombe, C., Catchen, M.D., Dansereau, G., Higino, G., Malpas, T., Mayall, H., Norman, K., Gravel, D., and Poisot, T. (2025). Deciphering probabilistic species interaction networks. Ecology Letters, 28(6), e70161. https://doi.org/10.1111/ele.70161 *2025*

[10] Strydom, T., Bouskila, S., **Banville, F.**, Barros, C., Caron, D., Farrell, M.J., Fortin, M.-J., Mercier, B., Pollock, L.J., Runghen, R., Dalla Riva, G.V., and Poisot. T. (2023). Graph embedding and transfer learning can help predict potential species interaction networks despite data limitations. Methods in Ecology and Evolution, 14(12). https://doi.org/10.1111/2041-210X.14228 *2023*

[9] **Banville, F.**, Gravel, D., and Poisot, T. (2023). What constrains food webs? A maximum entropy framework for predicting their structure with minimal biases. PLOS Computational Biology, 19(9), e1011458. https://doi.org/10.1371/journal.pcbi.1011458 *2023*

[8] Higino, G.T., **Banville, F.**, Dansereau, G., Forero-Muñoz, N.R., Windsor, F., and Poisot, T. (2023). Mismatch between IUCN range maps and species interactions data illustrated using the Serengeti food web. PeerJ, 11, e14620. https://doi.org/10.7717/peerj.14620 *2023*

[7] Lawlor, J., **Banville, F.**, Forero-Muñoz, N.R., Hébert, K., Martínez-Lanfranco, J.A., Rogy, P., and MacDonald, A.A.M. (2022). Ten simple rules for teaching yourself R. PLOS Computational Biology, 18(9), e1010372. https://doi.org/10.1371/journal.pcbi.1010372 *2022*

[6] Strydom, T., Bouskila, S., **Banville, F.**, Barros, C., Caron, D., Farrell, M.J., Fortin, M.-J., Hemming, V., Mercier, B., Pollock, L.J., Runghen, R., Dalla Riva, G.V., and Poisot, T. (2022). Food web reconstruction through phylogenetic transfer of low-rank network representation. Methods in Ecology and Evolution, 13(12), 2838-2849. https://doi.org/10.1111/2041-210X.13835 *2022*

[5] Strydom, T., Catchen, M.D., **Banville, F.**, Caron, D., Dansereau, G., Desjardins-Proulx, P., Forero-Muñoz, N.R., Higino, G., Mercier, B., Gonzalez, A., Gravel, D., Pollock, L., and Poisot, T. (2021). A roadmap towards predicting species interaction networks (across space and time). Philosophical Transactions of the Royal Society B: Biological Sciences, 376(1837), 20210063. https://doi.org/10.1098/rstb.2021.0063 *2021*

[4] Higino G., Forero-Muñoz, N.R., **Banville, F.**, Dansereau, G., and Poisot, T. (2021). Computers can help us find raccoons and other living creatures. Front. Young Minds. 9(595275). https://doi.org/10.3389/frym.2021.595275 *2021*

[3] **Banville, F.**, Vissault, S., and Poisot, T. (2021). Mangal.jl and EcologicalNetworks.jl: Two complementary packages for analyzing ecological networks in Julia. Journal of Open Source Software, 6(61), 2721. https://doi.org/10.21105/joss.02721 *2021*

[2] Dansereau, G., **Banville, F.**, Basque, E., MacDonald, A.A.M., and Poisot, T. (2020). [Re] Chaos in a three-species food chain. ReScience C, 6(3), 5. https://doi.org/10.5281/zenodo.4022518 *2020*

[1] MacDonald, A.A.M., **Banville, F.**, and Poisot, T. (2020). Revisiting the links-species scaling relationship in food webs. Patterns, 0(0). https://doi.org/10.1016/j.patter.2020.100079 *2020*

Preprints

[3] Catchen, M.D., **Banville, F.**, Blyth, P., Brimacombe, C., Dansereau, G., Higino, G., Malpas, T., Mayall, H., Norman, K., Strydom, T., Pollock, L.J., Gonzalez, A., and Poisot, T. (2026). Designing optimal biodiversity observation networks. EcoEvoRxiv. https://ecoevorxiv.org/repository/view/14980/ *2026*

[2] **Banville, F.**, Burnel, C., Carlson, C.J., Eiffener, E., Munoz Acevedo, G., Paz, A., and Poisot, T. (2026). The Global Biodiversity Framework supports global assessment of One Health actions. EcoEvoRxiv. https://ecoevorxiv.org/repository/view/13028/ *2026*

[1] Fuster-Calvo, A., Higino, G.T., Parent, C., Caron, D., **Banville, F.**, Massol, F., Blanchet, F.G., Hebert, K., Pollock, L., Maiorano, L., Guimaraes, P.R., Silva, P., Thuiller, W., and Gravel, D. (2025). Tracking community change via network coherence. bioRxiv. https://doi.org/10.1101/2025.11.04.686602 *2025*

PhD thesis

Banville, F. (2024). Vers une théorie de l'entropie maximale des réseaux trophiques. Papyrus, Université de Montréal. https://hdl.handle.net/1866/40637 *2024*

Other publications

Journal articles

[1] Boiteux, M., Di Bacco, K., **Banville, F.**, and Fortin, P. (2026). Les jeunes sont prêts pour l'approche Une seule santé. Qu'en est-il du Québec? *Le Devoir*, Opinion, Idées. https://www.ledevoir.com/opinion/idees/979739/jeunes-sont-prets-approche-seule-sante-est-il-quebec *2026*

Position papers

[1] Ayena, L., **Banville, F.**, Boiteux, M., Canonne, R., Clara Dackey, Y.F., Bernard, C., Claron, M., Danzabarm, A., Di Bacco, K., Diouf, M.B., Gottarelli, C., Hamza, H., Hounhouenou, C.M.O., Nanoukon-A., S.H., Gladys Ngono, E.A., and Talla Mba, M. (2026). La jeunesse au cœur d'Une seule santé. *IRD*, Position paper. https://tinyurl.com/3mru6vk5 — *2026*

Presentations

Organized sessions

[4] Groom, Q., Schwantes, C., **Banville, F.**, Hoyos, J., Marceló-Díaz, D., Upham, N., and Poisot, T. (2025, October 23). Biodiversity data for One Health: Integrating open data to address global health challenges [Symposium]. Living Data 2025, Bogota, Colombia. — *2025*

[3] Arce Plata, M.I., **Banville, F.**, Cruz Rodriguez, C., Dansereau, G., Forero-Muñoz, N.R., and Poisot, T. (2024, October 24). Connecting the dots: Bringing together scientists, policymakers, and practitioners to better implement biodiversity indicators [Side event]. GEO BON pavilion, 2024 United Nations Biodiversity Conference (COP16), Cali, Colombia. — *2024*

[2] Leroux, E., Mélançon, V., **Banville, F.**, Brémaud, J., Robitaille, F., and Gholamhosseini, M. (2023, March 16-17). Complexity matters: subjectivity as practice in contemporary biology [Conference]. 33rd Biology Symposium of the University of Montreal, Montreal, Qc, Canada. — *2023*

[1] Dansereau, G., **Banville, F.**, and Strydom, T. (2022, August 14-19). Space Oddity: Thinking about ecological networks across space [Inspire session]. ESA and CSEE Joint Meeting, Montreal, Qc, Canada. — *2022*

Invited talks

[1] Poisot, T., **Banville, F.**, Hall, I. and Samuels, A. (2025, November 14). Rethinking One Health through biodiversity [Panel discussion]. Symposium Une seule santé, GEO BON One Health Working Group session, Brossard, Canada. — *2025*

Contributed talks

[15] **Banville, F.**, Carlson, C., Eiffener, E., Munoz Acevedo, G., Paz Velez, A., and Poisot, T. (2026, February 24). Suivre la biodiversité pour protéger la santé humaine, animale et environnementale [Oral presentation]. 2026 QCBS Symposium, Montreal, Qc, Canada. — *2026*

[14] **Banville, F.** and Poisot, T. (2025, November 14). Relevance of biodiversity indicators for One Health action tracks [Oral presentation]. Symposium Une seule santé, GEO BON One Health Working Group session, Brossard, Canada. — *2025*

[13] **Banville, F.** and Poisot, T. (2025, October 23). Biodiversity monitoring and biosurveillance are fellow travellers [Oral presentation]. Living Data 2025, Bogota, Colombia. — *2025*

[12] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, October 23). Monitoring biodiversity for human, animal, and environmental health [Oral presentation]. Living Data 2025, Bogota, Colombia. — *2025*

[11] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, July 28). Monitoring biodiversity for human, animal, and environmental health [Webinar presentation]. GEO BON One Health Working Group Webinar: Towards biodiversity and health indicators, virtual. — *2025*

[10] **Banville, F.**, Gravel, D., and Poisot, T. (2025, July 6-9). Towards a maximum entropy theory of food webs [Conference presentation]. 2025 Annual Conference of the Canadian Society for Ecology and Evolution, Sherbrooke, Qc, Canada. — *2025*

[9] **Banville, F.**, Gravel, D., and Poisot, T. (2025, February 24-26). Vers une théorie de l'entropie maximale des réseaux trophiques [Conference presentation]. 2025 QCBS Symposium, Longueuil, Qc, Canada. *2025*

[8] **Banville, F.**, Gravel, D., and Poisot, T. (2024, March 21-22). Quoi, quand et où manger ? À la découverte des interactions trophiques entre contraintes et incertitudes [Conference presentation]. 34th Biology Symposium of the University of Montreal, Montreal, Qc, Canada. *2024*

[7] **Banville, F.**, Gravel, D., and Poisot, T. (2024, February 19-21). Quoi, quand et où manger ? À la découverte des interactions trophiques entre contraintes et incertitudes [Conference presentation]. 2024 QCBS Symposium, Montreal, Qc, Canada. *2024*

[6] **Banville, F.**, Gravel, D., and Poisot, T. (2023, May 8-12). Comment les réseaux de prédateurs et de proies sont-ils structurés dans les milieux naturels? [Conference presentation]. 90e congrès de l'ACFAS, Montreal, Qc, Canada. *2023*

[5] **Banville, F.**, Gravel, D., and Poisot, T. (2022, August 14-19). What constrains food webs? A maximum entropy model for predicting their structure with minimal biases [Conference presentation]. 2022 Annual Meeting of the Ecological Society of America (ESA), Montreal, Qc, Canada. *2022*

[4] **Banville, F.**, Gravel, D., and Poisot, T. (2022, March 25). Food webs of maximum entropy: A story of ecology and stochasticity [Conference presentation]. 32nd Biology Symposium of the University of Montreal, Montreal, Qc, Canada. *2022*

[3] **Banville, F.**, Gravel, D., and Poisot, T. (2020, October 22). Predicting networks of species interactions [Conference presentation]. IVADO Digital October 2020, virtual. *2020*

[2] **Banville, F.**, MacDonald, A.A.M., Gravel, D., and Poisot, T. (2020, February 19). How to estimate network structure without interaction data [Conference presentation]. Extreme Climate Events Symposium 2020, Toronto, On, Canada. *2020*

[1] **Banville, F.**, MacDonald, A.A.M., Gravel, D., and Poisot, T. (2019, December 18-20). How to estimate network structure without data [Conference presentation]. 10th Annual QCBS Symposium, Montreal, Qc, Canada. *2019*

Lightning talks

[5] **Banville, F.**, Hughes, A., Millette, K., and Poisot, T. (2026, April 6). Le Jenga de la biodiversité et de la santé [Conference short presentation]. La jeunesse interpelle le monde, One Health Summit, Lyon, France. *2026*

[4] **Banville, F.**, Gravel, D., and Poisot, T. (2022, March 29). Interactions entre espèces : une histoire d'écologie et de hasard [Conference short presentation]. My IVADO research project in 180 seconds, Montreal, Qc, Canada. *2022*

[3] **Banville, F.**, Gravel, D., and Poisot, T. (2021, October 28). Predicting food webs across space: First estimates of food-web structure derived from species richness [Conference short presentation]. IVADO Digital October 2021, virtual. *2021*

[2] **Banville, F.**, Gravel, D. and Poisot, T. (2020, December 14-16). Trophic-METE: A parsimonious theory of food-web structure [Conference short presentation]. 11th Annual QCBS Symposium, virtual. *2020*

[1] **Banville, F.**, Vissault, S., Bélisle, Z., Hoebeke, L., Stock, M., Szefer, P., and Poisot, T. (2020, July 29-31). Analyzing species interaction networks in Julia [Conference short presentation]. Juliacon 2020, virtual. *2020*

Posters

[3] **Banville, F.**, Carlson, C., Paz Velez, A., and Poisot, T. (2025, July 6-9). Monitoring biodiversity for human, animal, and environmental health [Poster presentation]. 2025 Annual Conference of the Canadian Society for Ecology and Evolution, Sherbrooke, Qc, Canada. *2025*

[2] **Banville, F.**, Gravel, D. and Poisot, T. (2021, December 8-10). Given limited knowledge, what can we say about a food web's properties? [Poster presentation]. 12th Annual QCBS Symposium, virtual. *2021*

[1] **Banville, F.**, Gravel, D. and Poisot, T. (2020, December 14-16). Trophic-METE: A parsimonious theory of food-web structure [Poster presentation]. 11th Annual QCBS Symposium, virtual. *2020*

PhD defense

Banville, F. (2024, December 2). Vers une théorie de l'entropie maximale des réseaux trophiques [PhD defense]. University of Montreal, Montreal, Qc, Canada. *2024*

Teaching and mentoring

Teaching assistance

F-2024	**Head teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
F-2024	**Teaching assistant**, BIO2043 Statistique pratique pour sciences de la vie	*U. de Montréal*
S-2024	**Head teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2024	**Head teaching assistant**, BIO2042 Biostatistique 2	*U. de Montréal*
F-2023	**Head teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
S-2023	**Teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2023	**Teaching assistant**, BIO2042 Biostatistique 2	*U. de Montréal*
F-2022	**Head teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
S-2022	**Teaching assistant**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2022	**Grader**, BIO2811 Dynamique des populations	*U. de Montréal*
F-2021	**Grader**, BIO1001 Méthodes de recherche en biologie (TP)	*U. de Montréal*
S-2021	**Teaching assistant**, BIO6065 École d'été en synthèse écologique de données	*U. de Montréal*
W-2021	**Grader**, BIO2811 Dynamique des populations	*U. de Montréal*
F-2020	**Teaching assistant**, BIO3043 Théorie des réseaux	*U. de Montréal*

Tutoring and mentoring

F-2024	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
S-2024	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2024	**Tutor**, BIO2042 Biostatistique 2	*U. de Montréal*
F-2023	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
S-2023	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2023	**Tutor**, BIO2042 Biostatistique 2	*U. de Montréal*
S-2022	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2022	**Mentor**, IVADO's Data.Trek training program	*virtual*
S-2021	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
S-2020	**Tutor**, BIO2041 Biostatistique 1	*U. de Montréal*
W-2020	**Mentor**, IVADO's Data.Trek training program	*virtual*

Workshop facilitation

S-2025	**Presentation assistant**, CSEE workshop, Data analysis and visualization in Python for ecologists (1 day)	*Sherbrooke, CAN*
W-2025	**Co-presenter**, BIOS^2 workshop, R and Git: From code to collaboration (1 day)	*Montreal, CAN*
S-2022	**Co-presenter**, Poisot lab Twitch workshop, Species distribution models in Julia (3 hours)	*virtual*
F-2021	**Co-presenter**, Software carpentry, Programming in R (4 days)	*Rimouski, CAN*
F-2021	**Presentation assistant**, Software carpentry, Introduction to Git (1 day)	*Rimouski, CAN*
W-2021	**Presentation assistant**, QCBS R workshops, Generalized linear models in R (3 hours)	*virtual*
W-2021	**Presentation assistant**, QCBS R workshops, Generalized additive models in R (3 hours)	*virtual*
W-2020	**Lead presenter**, IVADO's Data.Trek program, Machine learning and ecological networks in Julia (2 hours)	*Montreal, CAN*
W-2020	**Co-presenter**, IVADO's Data.Trek program, Introduction to R (3 hours)	*Montreal, CAN*
F-2019	**Lead presenter**, QCBS R workshops, Programming in R (3 hours)	*Montreal, CAN*

Internships

PhD internships

Research agent (internship 2)

Montreal, CAN

Group on Earth Observations Biodiversity Observation Network (GEO BON) — *May 2021 - Aug. 2021*

- forecasting beta diversity in Quebec and Colombia under different climate change scenarios
- supervision: Drs. Andrew Gonzalez (McGill University) and Timothée Poisot

Event co-organizer (internship 1)

Montreal, CAN

Institute for Data Valorization (IVADO) — *Jan. 2020 - Apr. 2020*

- organizing the Data.Trek training program
- supervision: Barbara Decelle

BSc internships - Biology

Research assistant (internship 3)

Montreal, CAN

Université de Montréal, département de sciences biologiques — *May 2019 - Aug. 2019*

- analyzing the impact of environmental variables on lakes' water layers in Quebec
- supervision: Drs. Jean- François Lapierre and Roxane Maranger

Research assistant (internship 2)

Montreal, CAN

Université de Montréal, département de sciences biologiques — *May 2018 - Aug. 2018*

- estimating mercury fluxes in Quebec rivers
- supervision: Drs. Jean- François Lapierre and Marc Amyot

Research assistant (internship 1)

Montreal, CAN

Institut de recherche en biologie végétale (IRBV) — *May 2017 - Aug. 2017*

- understanding the growth of potato ovules in vitro
- supervision: Drs. Daniel Philippe Matton and Valentin Joly

BSc internships - Mathematics

Research agent (internship 1)

Montreal, CAN

Collège des médecins du Québec — *Jan. 2015 - July 2015*

- analyzing the demography of medical doctors in Quebec and the quality of their work using survey results
- supervision: Johanne Thiffault

Working groups

WG participant

virtual

GEO BON One Health working group — *Jan. 2025 - May 2026*

- objective: mainstreaming environmental expertise in the One Health approach
- led by Timothée Poisot

WG co-organizer

Montreal, CAN

Computational Biodiversity Science and Services (BIOS^2) — *Sept. 2022*

- title: Black holes and revelations: Identifying priority sampling locations for local food webs in CAN
- led by Gabriel Dansereau, **Francis Banville**, Michael Catchen, and Tanya Strydom

WG participant

virtual

Canadian Institute of Ecology and Evolution — *Jan. 2021 - Mar. 2022*

- title: Assembling, predicting and refining a predator-prey metaweb for CAN
- led by Dominique Caron and Tanya Strydom

WG participant *virtual*

Living Data Project *Sept. 2021*

- title: Finding indicator species by assessing the utility of sampled abundance indices
- led by Robin Freeman, Jessica Currie, and Valentina Marconi

International delegations

Delegate to the One Health Summit *Lyon, France*

Les Offices jeunesse internationaux du Québec (LOJIQ) *Apr. 2026*

Delegate to the COP16 on biodiversity *Cali, Colombia*

Group on Earth Observations Biodiversity Observation Network (GEO BON) *Oct. 2024*

Services and outreach

Peer reviewer *Virtual*

Scientific journals *Jan. 2025 - present*

- Human and Ecological Risk Assessment: An International Journal (1 peer review)
- Oikos journal (1 peer review)
- Journal of Open Source Software (1 peer review)

Project participant *Saint-Hippolyte, CAN*

Blitz the Gap 2025 *July 2025*

- helped fill gaps in our knowledge of biodiversity in Canada

Volunteer *Longueuil, CAN*

Quebec Center for Biodiversity Science (QCBS) *Feb. 2025*

- provided help during the annual symposium, including the moderation of oral presentations

Speaker *Montreal, CAN*

La Nuit des chercheuses et des chercheurs, Espace pour la vie *Nov. 2021-23*

- presented my research one on one with the general public

Co-organizer of the annual symposium of the department of biological sciences *U. de Montréal*

Association des étudiants-chercheurs en biologie de l'Université de Montréal (AECBUM) *Aug. 2022 - Apr. 2023*

- organized the theme and talks of the symposium and the caterers

Student mentor *U. de Montréal*

Centre de l'engagement étudiant *Sept. 2022 - Dec. 2022*

- provided assistance to new students on campus

Environmental Coordinator *U. de Montréal*

Association étudiante de biologie de l'Université de Montréal (AEBUM) *Feb. 2016 - Aug. 2016*

- organized awareness-raising activities related to the environment

Treasurer *U. de Montréal*

Club Végé de l'Université de Montréal *Jan. 2015 - Aug. 2015*

- organized awareness-raising activities related to meat consumption and managed the club's finances

Completed graduate courses and certificates

Credited courses

W-2026	**PPA6617**, Gestion de classe en enseignement supérieur	*U. de Montréal*
F-2025	**PLU6035**, Enseigner au collégial aujourd'hui	*U. de Montréal*
S-2025	**PPA6075**, Apprentissage en enseignement supérieur	*U. de Montréal*
W-2025	**ETA6045**, Évaluation en enseignement supérieur	*U. de Montréal*
W-2021	**BIO860M**, Séminaire thématique en écologie	*UQAM*
F-2019	**BIO6037**, Analyse des réseaux écologiques	*U. de Montréal*
S-2019	**BIO6063**, Travail dirigé 1	*U. de Montréal*
S-2019	**BIO6065**, École d'été en synthèse écologique de données	*U. de Montréal*
W-2019	**BIO6032**, Biologie computationnelle et modélisation	*U. de Montréal*
W-2019	**BIO6033**, Méthodes quantitatives en biologie	*U. de Montréal*
W-2019	**BIO611**, Progrès en phylogénie systématique	*U. de Montréal*
W-2019	**MSO6028**, Introduction aux théories de la mesure	*U. de Montréal*
F-2018	**BIO6004**, Communication scientifique	*U. de Montréal*
F-2018	**BIO6077**, Analyse quantitative des données	*U. de Montréal*
F-2018	**BIO6260**, Génomique microbienne	*U. de Montréal*
F-2018	**GEO6321**, Travaux pratiques en géomatique	*U. de Montréal*

Extracurricular courses

S-2023	**EFI**, Short Course on Forecasting for Decision-Making: An Epidemiological and Ecological Perspective	*U. of Toronto*
S-2023	**ECL807**, Advanced Field School in Computational Ecology 2023	*U. de Sherbrooke*
F-2022	**LDP**, Scientific collaboration in ecology and evolution	*Living Data Project*
F-2021	**LDP**, Synthesis statistics for ecology and evolution	*Living Data Project*
S-2021	**ECL807**, École d'été en modélisation de la biodiversité 2021	*U. de Sherbrooke*

Extracurricular certificates

S-2024	**CIT**, Certified Carpentries Instructor	*The Carpentries*
W-2022	**LDP**, Cert. in Synthetic and Collaborative Science	*Living Data Project*

Affiliations and professional memberships

2025-26	**Postdoc member**, The Viral Emergence Research Initiative (Verena)	*USA*
2025-26	**Postdoc member**, Groupe de recherche en épidémiologie des zoonoses et santé publique (GREZOSP)	*U. de Montréal*
2024-26	**Certified Carpentries Instructor**, The Carpentries	*USA*
2023-26	**Student and postdoc member**, Group on Earth Observations Biodiversity Observation Network (GEO BON)	*CAN*
2019-26	**Student and postdoc member**, Quebec Centre for Biodiversity Science (QCBS)	*Quebec, CAN*
2019-26	**Lab member**, Quantitative and Computational Ecology Lab	*U. de Montréal*
2025	**Lab member**, The Carlson Lab	*Yale U.*
2019-25	**Fellow**, Computational Biodiversity Science & Services program (BIOS^2)	*CAN*
2019-24	**Lab member**, Integrative Ecology Lab	*U. de Sherbrooke*
2023	**Student member**, Ecological Forecasting Initiative (EFI)	*USA*
2023	**Student member**, Association canadienne-française pour l'avancement des sciences (ACFAS)	*Quebec, CAN*
2019-23	**Scholarship recipient**, Institute for Data Valorization (IVADO)	*Quebec, CAN*
2022	**Student member**, Ecological Society of America (ESA)	*USA*